PROJECT STORY

Rakuten product classification

Classify 84916 products into 27 categories from their title, description and image. All scores are weighted F1 on the same 16984 validation products.

> **Best model: a Random Forest that combines CamemBERT, a TF-IDF SVM and an image model, with 0.8988 weighted F1. Text carries most of the signal; images help on books.**

THE FINAL PIPELINE

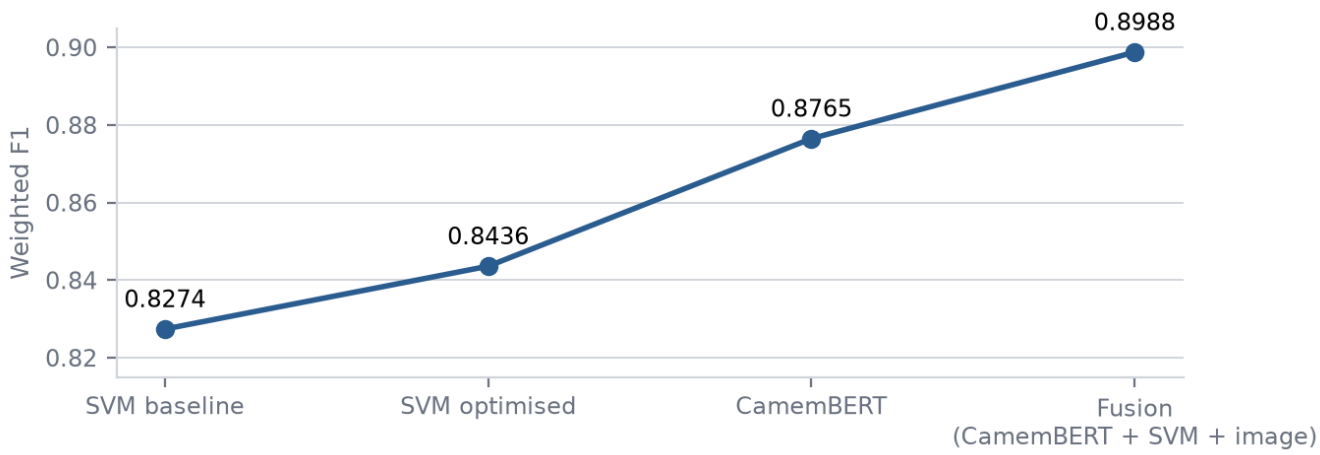
Title + description > TF-IDF + SVM 27 probabilities

Title + description > CamemBERT 27 probabilities

Product image > ConvNeXt-tiny 27 probabilities

81 probabilities > Random Forest > Category (F1 0.8988)

HOW THE SCORE IMPROVED

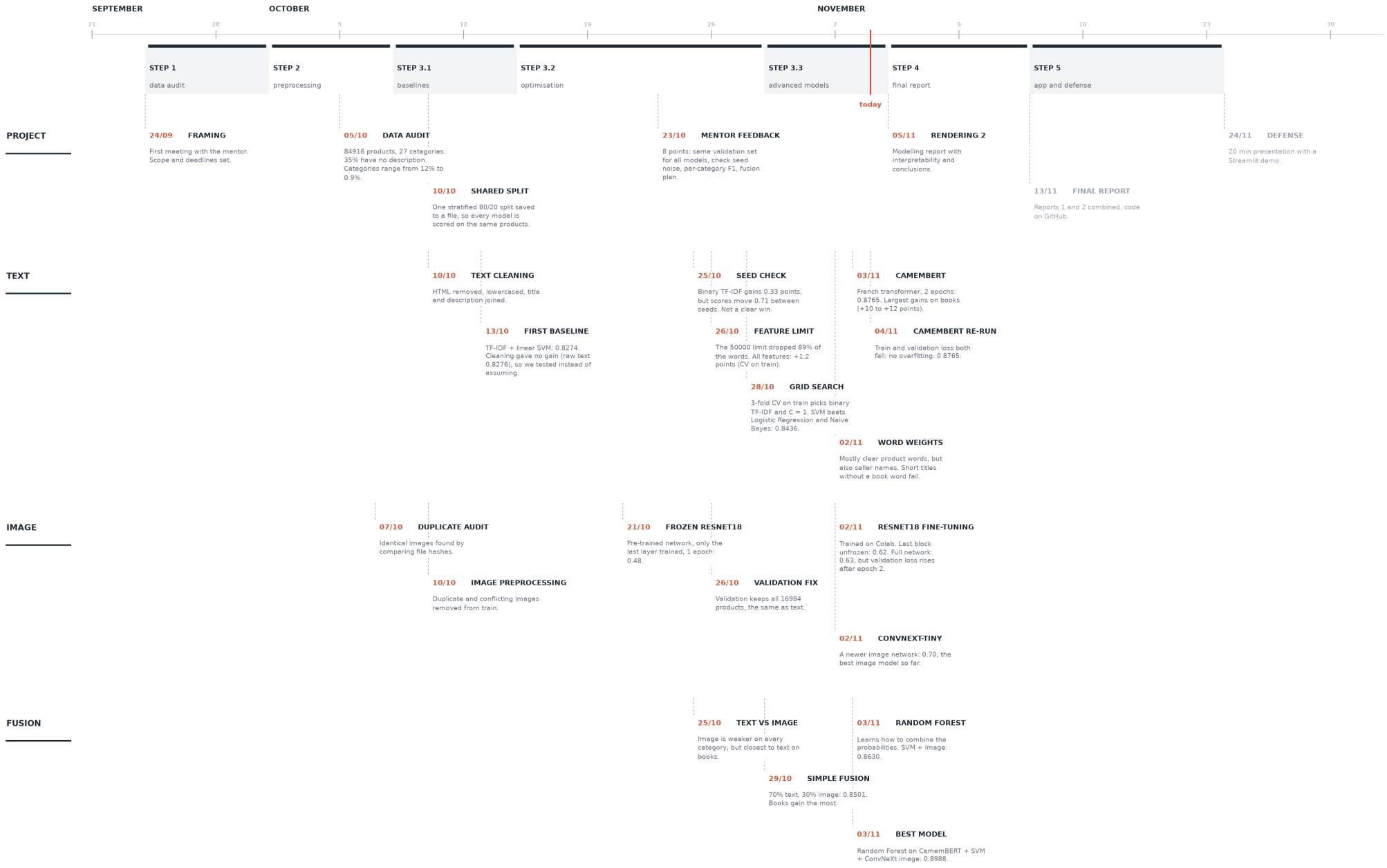


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7. Fusion	8. Interpretability	9. Conclusions

Rakuten product classification: project timeline

What we did, when, and why. Scores are weighted F1 on the same 16984 validation products.



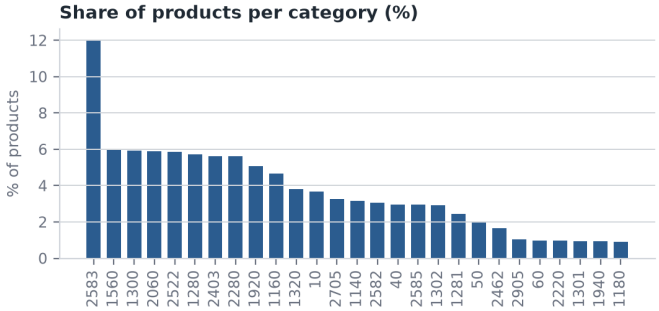
STEP 1 1 OCT

Data audit

> **The text is rich but messy, and the categories are unbalanced. We score with weighted F1 and also report macro F1.**

WHAT WE FOUND

- 84916 products, 27 categories, each with a title, an optional description and one image.
- 35% of products have no description. 18% of descriptions contain HTML.
- The largest category (2583, pool) has 12% of products, the smallest (1180) 0.9%: 13 times less.
- Category codes have no names. We named them by reading product titles.



Concept: weighted vs macro F1. F1 combines precision (how many predicted products are right) and recall (how many true products are found). **Weighted F1** averages the 27 category scores weighted by their size: it is the challenge metric. **Macro F1** gives every category the same weight, so it shows if small categories are ignored.

EXAMINER QUESTIONS

Why not accuracy?

With unbalanced categories, a model can ignore small ones and still look good. F1 per category, averaged, makes this visible.

Is the imbalance a problem for training?

Not much here: the smallest category still has 764 products. We check it with macro F1 and per-category F1 instead of re-sampling.

STEP 2 8 OCT

Preprocessing and a shared split

> **Every model is scored on the same 16984 validation products. Without this, text, image and fusion scores could not be compared or combined.**

TEXT

- HTML removed, text lowercased.
- Title and description joined into one text.

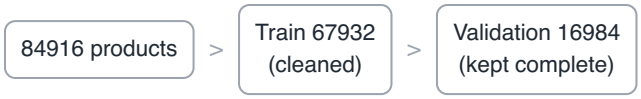
IMAGES

- Duplicate images found by comparing a hash (fingerprint) of each file.
- 5620 duplicate or conflicting images removed from **train only**.

SPLIT

- One stratified 80/20 split: 67932 train, 16984 validation, saved to a file.
- Stratified means each category has the same share in both parts.

Concept: why validation must stay untouched. The validation set plays the role of new, unseen products. If we clean it, remove rows from it, or tune settings on it, its score becomes too optimistic. So all cleaning that removes products is done on train only.



The mentor found that an early version of the image cleaning also removed 551 validation products. This was fixed: the image model now uses the same 16984 products as text.

EXAMINER QUESTIONS

Why keep duplicate images in validation?

At test time every product needs a prediction, duplicate or not. Validation must look like the real test.

Did the text cleaning help?

No: 0.8276 on raw text against 0.8274 on clean text. We measured it instead of assuming it. TF-IDF already gives little weight to rare HTML tags.

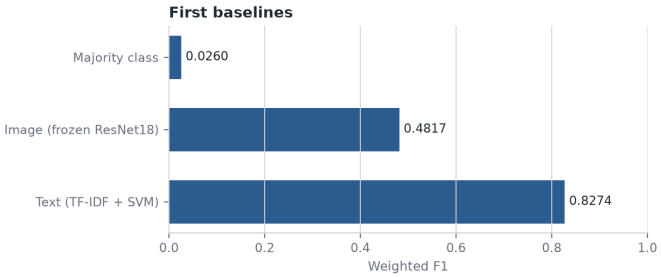
STEP 3.1 15 OCT

Baselines

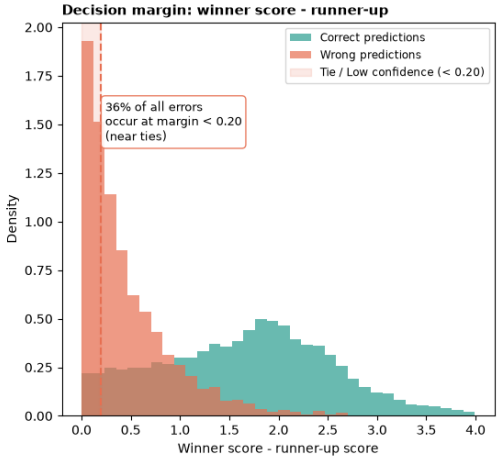
> **Text is far stronger than images. A simple TF-IDF + SVM already reaches 0.83, the image model 0.48.**

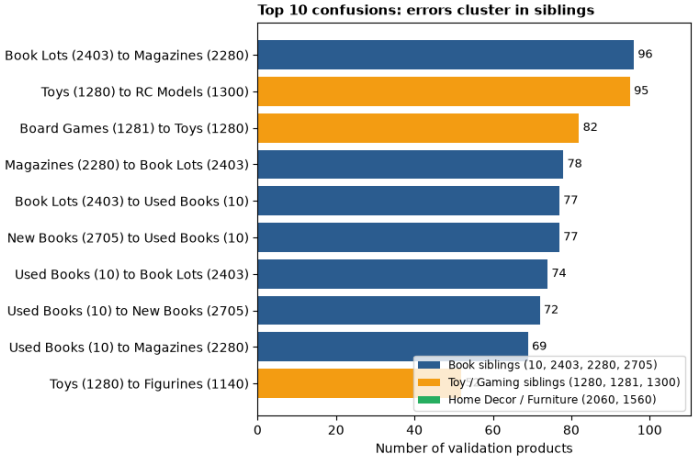
Concept: TF-IDF. Each product becomes a long list of numbers, one per word or word pair. A word gets a high value if it appears in this product but is rare overall (*piscine*), and a low value if it appears everywhere (*pour*).

Concept: linear SVM. For each category, the model learns one weight per word. A product's score for a category is the sum of its word values times the weights. The highest score wins.



Two ideas did not clearly help: cleaning the text (0.8276 raw, 0.8274 clean) and binary TF-IDF with 50000 features (+0.33 on average over 5 splits, while the score moves 0.71 between splits).





EXAMINER QUESTIONS

Is 0.83 better than the official benchmark (0.8113)?

Not directly comparable: the benchmark is a CNN on titles only, scored on Rakuten's hidden test set. Our reference is our own baseline on our validation.

Where does the SVM fail?

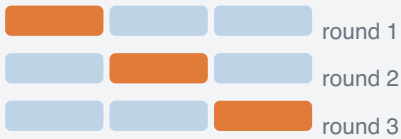
36% of errors have a margin under 0.20: the model hesitated between two categories. 7 of the 10 most confused pairs are book categories.

STEP 3.2 29 OCT

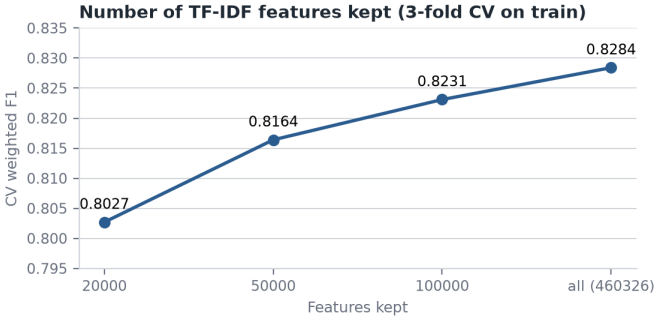
Optimisation of the text model

> **Using all words instead of 50000 and tuning with cross-validation on train: 0.8313 to 0.8436.**

Concept: 3-fold cross-validation. Train is cut into 3 parts. The model trains on 2 and is tested on the third, 3 times. The average is used to choose settings, so validation is never used for choices.

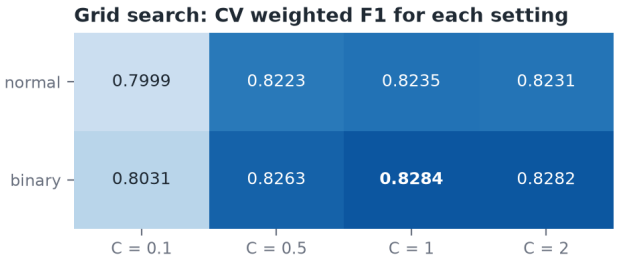


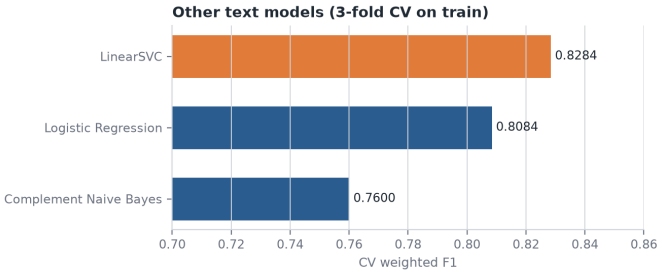
blue: train, orange: test



The 50000 limit kept only the most frequent words and dropped 89% of them, including rare but useful ones such as brand or model names.

Concept: C in the SVM. C sets how strictly the SVM fits the training data. Small C gives a simpler model (strong regularisation), large C fits every training product, including noise.





EXAMINER QUESTIONS

Was validation used to choose settings?

No. Feature limit, binary and C were chosen with 3-fold CV on train. Validation was scored once at the end.

Why keep binary TF-IDF now?

With all features its gain (0.50 points) is larger than the CV spread (0.30). We set this rule before looking at the result.

Why LinearSVC and not Logistic Regression?

2 points better in CV, and still trains in seconds.

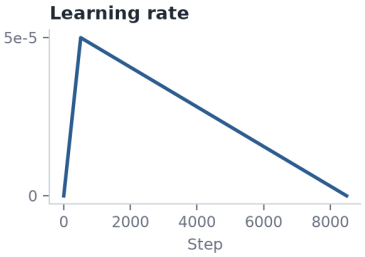
STEP 3.3 5 NOV

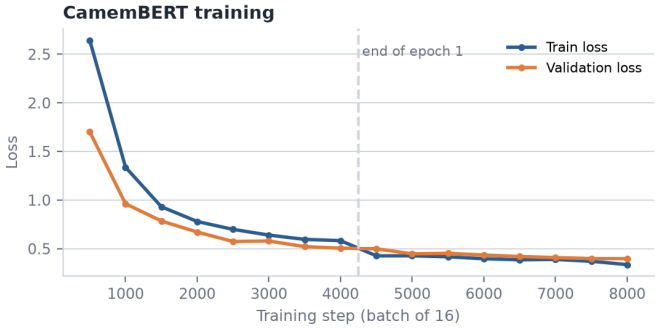
Deep learning for text: CamemBERT

> **CamemBERT reads words in context and reaches 0.8765, +3.4 points over the SVM. The largest gains are on the book categories.**

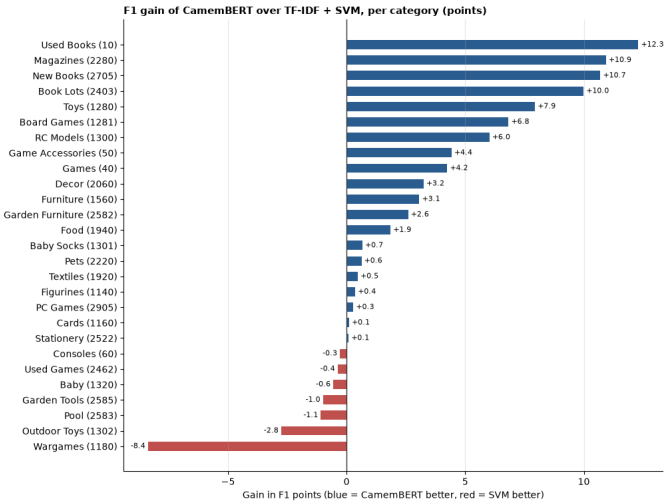
Concept: transformer and fine-tuning. CamemBERT is a transformer pre-trained on a large amount of French text, so it already knows French. Each word is read together with the words around it, so *lot de 5 tomes* and *lot de jardin* are understood differently. Fine-tuning adds a small layer for our 27 categories and trains the whole model a little more on our products.

Setting	Value
Input	title + description, 128 tokens
Epochs, batch size	2, 16
Optimiser	AdamW, learning rate 0.00005
Regularisation	weight decay 0.01, dropout 0.1
Stability	warmup, linear decay, gradient clipping 1.0





Validation loss keeps falling in both epochs (0.501, then 0.427): no overfitting. In epoch 2 train loss drops a little below it, as each product is seen a second time.



EXAMINER QUESTIONS

Why no cross-validation for CamemBERT?

One training run takes about 2 hours on our laptop. We used the standard settings and tuned nothing, so validation only measures the result.

Why is it better than TF-IDF?

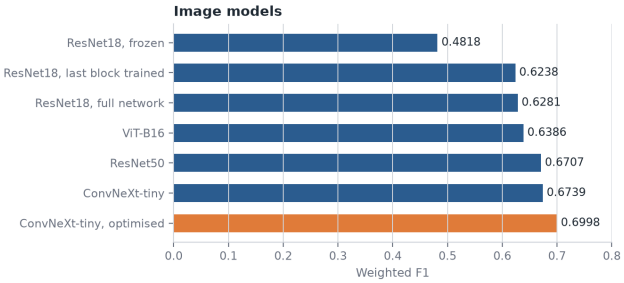
TF-IDF loses word order and context. Book titles share the same words (tome, édition) but not the same context.

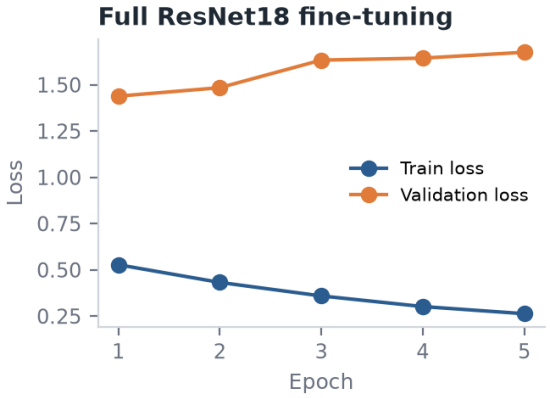
STEP 3.3 5 NOV

Image models

> **Images alone are weak (0.48 frozen, 0.70 with ConvNeXt), but their errors differ from text.**

Concept: frozen vs fine-tuned network. A pre-trained image network already recognises shapes and objects. **Frozen**: only the last layer is trained, fast but limited. **Fine-tuned**: deeper layers are also trained, so the network adapts to product photos.





When the whole network is trained, train loss keeps falling but validation loss rises after epoch 1: the model starts to memorise the training images.

EXAMINER QUESTIONS

Why was train loss higher than validation loss for the frozen model?

Train loss is averaged over the epoch while the model is still learning, and validation loss is measured at the end. With one epoch this gap is expected.

Why use a model that only scores 0.48 to 0.70?

Because it fails on different products than text. Combined, it fixes text errors, mainly on books.

STEP 3.3 5 NOV

Fusion: combining text and image

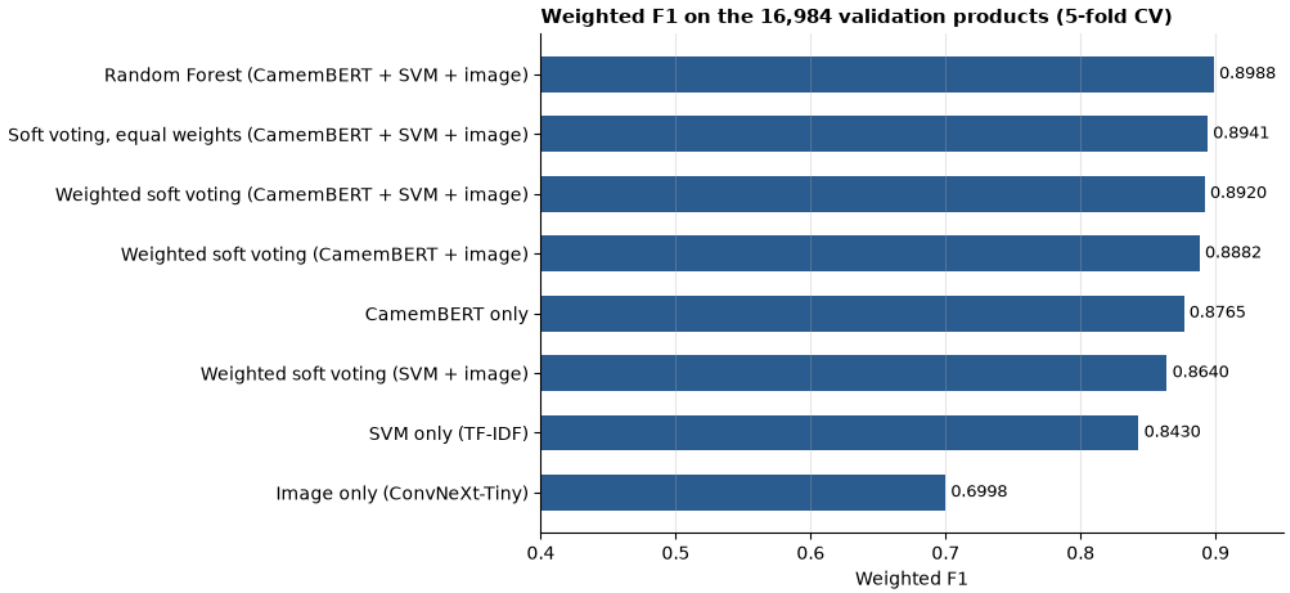
> **A Random Forest that learns how to combine the models beats any single model: 0.8988.**

SVM 27 probs > CamemBERT 27 probs > Image 27 probs > Random Forest 81 inputs > Category

Concept: stacking. Each model gives 27 probabilities per product. A second model, here a Random Forest, takes these 81 numbers as input and learns when to trust which model, for example "for books, listen to the image more".

Concept: calibration. An SVM gives scores, not probabilities. Calibration learns on the training data how to turn these scores into probabilities, so they can be combined with the other models.

Concept: Random Forest. It trains 300 decision trees on random samples and lets them vote (bagging). We also tried Gradient Boosting (trees trained one after another, each fixing the previous errors): slightly lower and much slower.



EXAMINER QUESTIONS

Is the fusion score optimistic?

No. The Random Forest is scored with 5-fold cross-validation on the validation products: every product is predicted by a model that never saw it.

Why keep the SVM if CamemBERT is better?

They make different errors: CamemBERT reads context, the SVM catches rare exact words. Wargames, CamemBERT's weakest category, gains 7.8 points when both are combined.

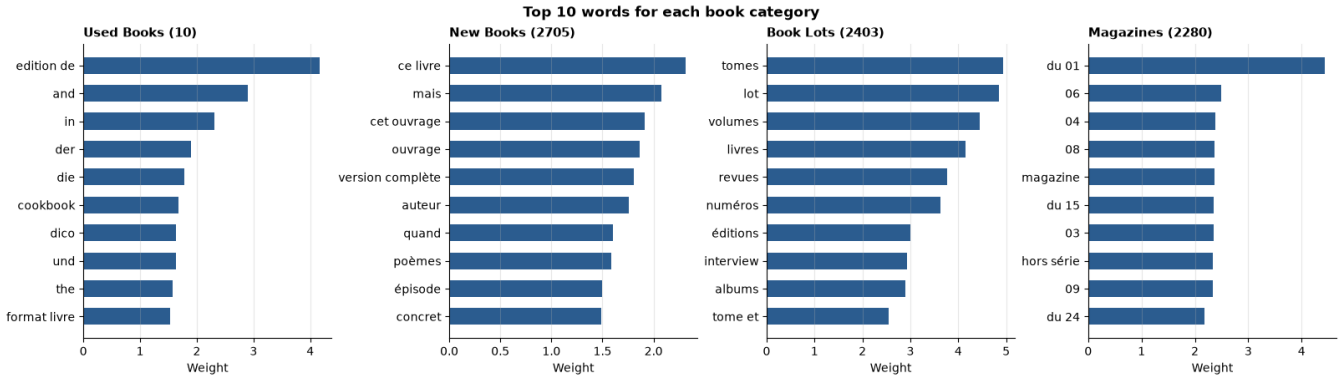
STEP 3.3 5 NOV

Interpretability

> **The text model mostly uses sensible product words, but it also learned seller names and writing style.**

Concept: reading the SVM weights. A linear SVM has one weight per word and category. The highest weights show which words push a product toward a category.

- Clear words: *piscine* for pool, *chien* for pets, *warhammer* for wargames.
- Seller names: *sweetheart93*, *germenly*. This works on this data but may not on new sellers.
- Writing style: new books use summary words (*cet ouvrage*), used books often have English or German titles.
- Short titles with no book word (*la promenade au parc*) give the model nothing to decide with.



EXAMINER QUESTIONS

Can the model be trusted on new sellers?

Partly. Part of its score comes from seller names. It should be monitored on new sellers and retrained regularly.

CONCLUSIONS

What we learned

SCIENTIFIC

- Text carries most of the signal. Images alone are weak but help where text is weak.
- Context matters: CamemBERT beats TF-IDF by 3.4 points, mostly on books.
- A learned combination beats the best soft voting (equal weights): 0.8988 against 0.8941.
- Several "obvious" ideas did not help when measured: text cleaning, and binary TF-IDF with 50000 features.

BUSINESS

- About 9 in 10 products are put in the right category automatically.
- Low-confidence products, mainly books, should go to a human or ask the seller to confirm.
- Requiring a description would help: short titles are the main source of errors.

LIMITS

- Scores are on our validation split, not on Rakuten's hidden test set.
- CamemBERT settings were not tuned, because each run takes 2 hours.
- Category names are our guesses; Rakuten does not publish them.
- Part of the text model relies on seller names.

NEXT STEPS

- Image ensemble (0.72): check how it was built, then test it in the fusion.
- Train CamemBERT for a third epoch and on longer texts.
- Streamlit app with confidence levels for each prediction.